



Cochrane
Library

Cochrane Database of Systematic Reviews

Creatine for women in pregnancy for neuroprotection of the fetus (Review)

Dickinson H, Bain E, Wilkinson D, Middleton P, Crowther CA, Walker DW

Dickinson H, Bain E, Wilkinson D, Middleton P, Crowther CA, Walker DW.
Creatine for women in pregnancy for neuroprotection of the fetus.
Cochrane Database of Systematic Reviews 2014, Issue 12. Art. No.: CD010846.
DOI: [10.1002/14651858.CD010846.pub2](https://doi.org/10.1002/14651858.CD010846.pub2).

www.cochranelibrary.com

TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	2
BACKGROUND	3
OBJECTIVES	5
METHODS	5
RESULTS	6
DISCUSSION	6
AUTHORS' CONCLUSIONS	7
ACKNOWLEDGEMENTS	7
REFERENCES	8
APPENDICES	10
CONTRIBUTIONS OF AUTHORS	13
DECLARATIONS OF INTEREST	13
SOURCES OF SUPPORT	14
DIFFERENCES BETWEEN PROTOCOL AND REVIEW	14
INDEX TERMS	14

[Intervention Review]

Creatine for women in pregnancy for neuroprotection of the fetus

Hayley Dickinson¹, Emily Bain², Dominic Wilkinson³, Philippa Middleton², Caroline A Crowther^{2,4}, David W Walker¹

¹The Ritchie Centre, MIMR-PHI Institute of Medical Research, Melbourne, Australia. ²ARCH: Australian Research Centre for Health of Women and Babies, Robinson Research Institute, Discipline of Obstetrics and Gynaecology, The University of Adelaide, Adelaide, Australia. ³Oxford Uehiro Centre for Practical Ethics, University of Oxford, Oxford, UK. ⁴Liggins Institute, The University of Auckland, Auckland, New Zealand

Contact: Dominic Wilkinson, Oxford Uehiro Centre for Practical Ethics, University of Oxford, Oxford, UK.
dominic.wilkinson@philosophy.ox.ac.uk.

Editorial group: Cochrane Pregnancy and Childbirth Group.
Publication status and date: New, published in Issue 12, 2014.

Citation: Dickinson H, Bain E, Wilkinson D, Middleton P, Crowther CA, Walker DW. Creatine for women in pregnancy for neuroprotection of the fetus. *Cochrane Database of Systematic Reviews* 2014, Issue 12. Art. No.: CD010846. DOI: [10.1002/14651858.CD010846.pub2](https://doi.org/10.1002/14651858.CD010846.pub2).

Copyright © 2014 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

ABSTRACT

Background

Creatine is an amino acid derivative and, when phosphorylated (phosphocreatine), is involved in replenishing adenosine triphosphate (ATP) via the creatine kinase reaction. Cells obtain creatine from a diet rich in fish, meat, or dairy and by endogenous synthesis from the amino acids arginine, glycine, and methionine in an approximate 50:50 ratio. Animal studies have shown that creatine may provide fetal neuroprotection when given to the mother through her diet in pregnancy. It is important to assess whether maternally administered creatine in human pregnancy (at times of known, suspected, or potential fetal compromise) may offer neuroprotection to the fetus and may accordingly reduce the risk of adverse neurodevelopmental outcomes, such as cerebral palsy and associated impairments and disabilities arising from fetal brain injury.

Objectives

To assess the effects of creatine when used for neuroprotection of the fetus.

Search methods

We searched the Cochrane Pregnancy and Childbirth Group's Trials Register (30 November 2014).

Selection criteria

We planned to include all published, unpublished, and ongoing randomised trials and quasi-randomised trials. We planned to include studies reported as abstracts only as well as full-text manuscripts. Trials using a cross-over or cluster-randomised design were not eligible for inclusion.

We planned to include trials comparing creatine given to women in pregnancy for fetal neuroprotection (regardless of the route, timing, dose, or duration of administration) with placebo, no treatment, or with an alternative agent aimed at providing fetal neuroprotection. We also planned to include comparisons of different regimens for administration of creatine.

Data collection and analysis

We identified no completed or ongoing randomised controlled trials.

Main results

We found no randomised controlled trials for inclusion in this review.

Authors' conclusions

As we did not identify any randomised controlled trials for inclusion in this review, we are unable to comment on implications for practice. Although evidence from animal studies has supported a fetal neuroprotective role for creatine when administered to the mother during pregnancy, no trials assessing creatine in pregnant women for fetal neuroprotection have been published to date. If creatine is established as safe for the mother and her fetus, research efforts should first be directed towards randomised trials comparing creatine with either no intervention (ideally using a placebo), or with alternative agents aimed at providing fetal neuroprotection (including magnesium sulphate for the very preterm infant). If appropriate, these trials should then be followed by studies comparing different creatine regimens (dosage and duration of exposure). Such trials should be high quality and adequately powered to evaluate maternal and infant short and longer-term outcomes (including neurodevelopmental disabilities such as cerebral palsy), and should consider utilisation/costs of health care.

PLAIN LANGUAGE SUMMARY

Creatine for women in pregnancy for neuroprotection of the fetus

This review did not find any randomised controlled trials that looked at whether creatine, given to a mother in pregnancy, can help protect her baby's brain.

The developing fetal brain is very vulnerable to injury, which may arise from infection in the uterus, insufficient blood flow to the placenta, and long-term reduced oxygen in the baby's blood. Damage to the developing brain during pregnancy can lead to death of the baby, or, if the baby survives, to life-long problems such as hearing, sight and speech disorders, intellectual disability, and cerebral palsy.

Creatine is involved with cellular energy production and how energy is stored for use in the body's tissue. Its primary function is to regenerate adenosine diphosphate (ADP) to adenosine triphosphate (ATP) in body tissues with high and fluctuating energy demands. Adults obtain approximately half of their daily requirement of creatine from a diet containing fresh fish, meat, and other dairy products. The body makes the remainder of the creatine from amino acids (the building blocks of proteins). Experiments in animals have suggested that creatine might be able to protect the developing fetal brain from injury when given to the mother during pregnancy. Human studies of creatine, outside of pregnancy (such as in children following traumatic brain injury, and in adults with neurodegenerative conditions), have been promising, suggesting creatine may be able to protect the brain, and these studies have been reassuring, with an absence of any detected harm.

We found no completed (or ongoing) randomised controlled trials that assessed whether creatine given to the mother at times of known, suspected, or potential fetal compromise during pregnancy helps to protect the baby's brain. Randomised controlled trials are needed to establish whether creatine can protect against brain injury for the baby in the womb. The babies in these trials need to be followed up over a long period so that we can monitor the effects of creatine on their development into childhood and adulthood.

BACKGROUND

Description of the condition

Fetal brain injury: causes and consequences

The developing fetal brain is vulnerable to damage arising from hypoxia, infection/inflammation, and release of excitatory amino acids, and thus compromise of placental perfusion (via uterine or umbilical blood flow), trans-placental oxygen delivery, or increased pro-inflammatory cytokines in the intrauterine environment, increases the risk of brain injury (and/or abnormal brain development) for both the preterm (before 37 weeks' gestation) and term fetus (Rees 2011). Fetal brain injury is a major contributor to perinatal mortality and morbidity worldwide (Jensen 2003), with such injury being associated with a spectrum of life-long functional and behavioural disorders.

Injury to both the preterm and term developing brain is known to be associated with life-long and devastating sequelae, such as hearing, sight and speech disorders, seizures, intellectual disability, and motor impairments that may manifest as cerebral palsy (Vexler 2001). Cerebral palsy is an umbrella term, describing "*a group of disorders of the development of movement and posture, causing activity limitations, which are attributed to non progressive disturbances that occurred in the developing fetal or infant brain*" (Bax 2005). Cerebral palsy is a complex neurological condition, and is often found alongside cognitive, communication, sight and hearing impairments, or epilepsy, pain, behaviour, and sleep disorders (Novak 2012). It is the most common physical disability in childhood, and the most severe physical disability within the spectrum of developmental delay. While for a small number of individuals brain injury acquired after birth may lead to the development of cerebral palsy, for the vast majority (94%) with cerebral palsy, the injury leading to this condition occurs to the fetal brain *in utero* or to the infant brain before one month of age (ACPR Group 2009).

While a number of causes of fetal brain injury have been recognised (such as intrauterine infection, placental insufficiency, and chronic fetal hypoxia leading to metabolic derangement), episodes of cerebral hypoxia-ischaemia (reduced oxygen in the blood combined with reduced blood flow to the brain) appear to be important in a great number of cases (whether being acute, chronic, associated with inflammation, or as an antecedent of preterm birth) (du Plessis 2002; Rees 2011; Volpe 2001). Similarly, a great number of potential predisposing factors and causal pathways for cerebral palsy and associated impairments and disabilities have been identified. While it has been shown that neuronal cell injury predominates in term infants, and cerebral white matter injury predominates in premature infants (Volpe 2001), recent evidence suggests that white matter injury is also present in term infants, and grey matter injury in preterm infants (Rees 2011).

Though preterm birth has been recognised as one of the most important risk factors for cerebral palsy (Blair 2006; Jacobsson 2002; McIntyre 2013) (with preterm infants being at an increased risk of white matter injury such as periventricular leukomalacia, and of intraventricular haemorrhage (Larroque 2003)), approximately 60% of all children with cerebral palsy are born at term (ACPR Group 2009; McIntyre 2013; Wu 2003). For infants born at term, antenatal or intrapartum risk factors for cerebral palsy consistently identified in the literature have

included small-for-gestational age, low birthweight, and placental abnormalities (Blair 2006; McIntyre 2013). Maternal bleeding in the second and third trimesters (McIntyre 2013), hypertension in pregnancy (McIntyre 2013), pre-eclampsia (Blair 2006; McIntyre 2013), perinatal infection (such as chorioamnionitis) (Blair 2006; McIntyre 2013; Wu 2003), and increasing fetal plurality (Blair 2006) have each been shown to increase the risk of cerebral palsy and associated neurosensory disorders across all gestational ages. For term infants, intrapartum birth asphyxia (a condition resulting from deprivation of oxygen to a newborn, lasting long enough to cause physical harm) has also been shown to be an important predictor of brain injury and later disability (Dilenge 2001; McIntyre 2013).

Following cerebral hypoxia and ischaemia, it is believed that a sequence of pathophysiological events ultimately leading to cell death (via necrosis or apoptosis) are triggered, involving for example, the overstimulation of N-methyl-D-aspartate (NMDA) type glutamate receptors, the accumulation of calcium in cells, and the activation of deleterious events mediated by calcium (including the stimulation of enzymes such as nitric oxide synthase, and the production of oxygen free radicals) (Jensen 2003; Johnston 2000; Rees 2011). Studies of the developing fetal brain have shown that the nature and severity of insult, and gestational age at the time of injury, can greatly influence the subsequent neuropathology. An important common feature of the fetal brain in all such situations, however, is the depletion of cellular energy.

To date, there is minimal knowledge regarding effective strategies to prevent, reduce, or remove the risk of antenatally acquired fetal brain injury and, accordingly, prevent the potentially devastating life-long consequences for the infant, child, and adult. Magnesium sulphate, when given to the mother prior to very preterm birth, is one of the first antenatal interventions shown to be effective in reducing the risk of death and cerebral palsy for the infant (Doyle 2009). While the precise mechanism of action of magnesium sulphate for neuroprotection of the fetus is not known, experimental evidence and animal studies support several possible neuroprotective effects, for example, magnesium has been shown to prevent excitotoxic calcium-induced cell injury, through non-competitive voltage-dependent inhibition of the NMDA receptor to glutamate (thereby reducing calcium influx) (Marret 2007). In the Doyle 2009 Cochrane review, magnesium sulphate, when administered for the mother prior to preterm birth, was associated with a 32% relative reduction in the risk of cerebral palsy (risk ratio (RR) 0.68, 95% confidence interval (CI) 0.54 to 0.87; five trials; 6145 infants), with 63 babies needing to be treated to benefit one baby by avoiding cerebral palsy, and 42 babies treated to benefit one baby by avoiding death or cerebral palsy (Doyle 2009). While the benefits of this therapy for preterm infants were established in this Cochrane review, not all infants exposed to therapy showed improved outcomes (the absolute risk of cerebral palsy for infants exposed to antenatal magnesium sulphate was 3.4% and 5.0% for infants unexposed) (Doyle 2009). Currently, there is insufficient evidence to assess the efficacy and safety of magnesium sulphate when administered to women for neuroprotection of the term fetus (Nguyen 2013), and there are potential (though commonly minor) adverse effects for the mother associated with this treatment (Bain 2013). At present, other agents being investigated for providing antenatal fetal neuroprotection include maternally administered melatonin (Wilkinson 2013) and allopurinol (Kaandorp 2010; Kaandorp 2012); and while it has previously been shown that antenatal corticosteroids, when given

prior to preterm birth, can reduce the risk of cerebroventricular haemorrhage, respiratory distress, necrotising enterocolitis, and death for the neonate, the evidence for benefits into childhood, including reductions in neurodevelopmental delay and cerebral palsy, are less clear (Roberts 2006).

Following recent advances in understanding the mechanisms of fetal brain injury and in identifying predisposing factors, further promise has been raised for the development of primary preventative strategies, based on preventing the complex sequence of pathophysiological and biochemical events that induce irreversible injury. Ideally, a primary preventative agent would be cost-effective (and/or inexpensive), have a low potential for toxicity, be easily administered to women either in the inpatient or outpatient setting (i.e. available to those with low obstetric monitoring), and be broadly applicable, such that it may offer protection to both the preterm and near-term fetal brain in a range of obstetric situations, including known or suspected maternal/fetal compromise.

Description of the intervention

Creatine

Creatine is a simple guanidine compound, which may be synthesised endogenously from the amino acids arginine, glycine, and methionine, in the liver, kidney, and pancreas (Adcock 2002). It may also be ingested, through the consumption of dairy, fish, and meat, and is found throughout the human body, including in the brain (Rees 2011). Creatine is taken up into tissues via the creatine transporter and stored as creatine or phosphocreatine. Phosphocreatine is readily converted to creatine via creatine kinase, in a reversible reaction that yields a high energy phosphate allowing the conversion of adenosine diphosphate (ADP) to adenosine triphosphate (ATP) (Wallimann 1992).

A number of studies have demonstrated that creatine has neuroprotective and antioxidant properties, suggesting benefits for neurodegenerative diseases, including amyotrophic lateral sclerosis and Parkinson's disease, traumatic brain disease, and adult stroke; conditions encompassing hypoxia and excitotoxic-mediated brain injury (Sullivan 2000; Zhu 2004). A Cochrane review that included three trials assessing creatine for improving amyotrophic lateral sclerosis survival, or for slowing progression, found no clear evidence to support meaningful improvements. Importantly, however, creatine was found to be well tolerated, with no serious adverse effects observed (Pastula 2012). A new Cochrane review will assess the efficacy and safety of creatine when used alone, or as an adjunctive treatment, for Parkinson's disease (Wang 2012).

How the intervention might work

Creatine for fetal neuroprotection

There is currently increasing support for the use of creatine as a therapy for protecting tissues against injury; particularly, there is growing evidence of creatine's potential to act as a neuroprotective agent (Wallimann 2011). One of the primary mechanisms of injury arising from severe hypoxia at birth (particularly for the brain) involves mitochondrial dysfunction, leading to impaired energy metabolism and oxidative stress (Calvert 2005; Wyss 2002). It has been suggested that preservation of ATP through increase of the intracellular pool of creatine and phosphocreatine can

protect the brain from such injuries (Beal 2011; Wallimann 1992). In addition to its role as an 'energy buffer' (providing energy in the absence of oxygen), creatine appears to have antioxidant properties, scavenging free radicals (Lawler 2002; Sestili 2006). Creatine has also been shown to improve the recovery of cerebral blood flow following the cessation of a hypoxic episode (Prass 2006).

Hypoxic-ischaemic models of neonatal brain damage in rodents have provided support for the neuroprotective effects of creatine. Subcutaneous injections of creatine given to neonatal rodents prior to transient severe hypoxia-ischaemia have been shown to reduce brain oedema (Adcock 2002). Recently, the supplementation of the maternal diet with creatine, from mid-pregnancy until term, has been shown to not only increase the concentration of creatine and phosphocreatine in rodent fetal tissues, but also to improve survival and postnatal growth of the offspring after an acute hypoxic episode at birth (Ireland 2008). Maternal creatine supplementation during pregnancy has been shown to prevent lipid peroxidation and apoptosis in the brains of rodent offspring following intrapartum hypoxia (Ireland 2011). It has been proposed that creatine's ability to protect mitochondrial function may account for this observed neuroprotective effect (Ireland 2011).

In addition to offering neuroprotection, maternal creatine supplementation has been shown to protect the newborn diaphragm from intrapartum hypoxia-induced damage (Cannata 2010). Rodent offspring born to mothers who received creatine supplementation from mid-pregnancy have been shown to be less likely to incur diaphragmatic damage (including muscular atrophy and contractile dysfunction) following hypoxia, as compared with control offspring (Cannata 2010). Most recently, maternal creatine supplementation has been shown to protect the newborn kidney from intrapartum hypoxia-induced damage. Specifically, creatine given to the mother throughout the second half of pregnancy has been shown to be able to prevent structural damage to the glomeruli and tubules of the kidney of the newborn spiny mouse (Ellery 2013).

Importantly, as with any intervention during pregnancy, the impact for the mother must be considered, along with the impact on the normal development of the fetus. Studies have recently assessed the impact of maternal creatine supplementation from mid-gestation on the capacity for creatine synthesis and transport in the newborn spiny mouse; encouragingly, long-term supplementation was not shown to impact on the normal development of these pathways (Dickinson 2013). Similarly, to date, no effects of maternal creatine supplementation on maternal body composition have been observed when creatine-fed pregnant spiny mice have been compared with control-fed spiny mice (unpublished observations; Dickinson/Walker laboratory, manuscript under review). While there has been some concern over possible deleterious effects of long-term, high-dose creatine supplementation on kidney function, recent work, measuring chromium-ethylenediamine tetraacetic acid (51-Cr-EDTA) clearance, has indicated no negative impact of creatine supplementation on kidney function in human type 2 diabetic patients (Gualano 2011). Studies measuring urine creatinine (as compared with 51-Cr-EDTA), as a marker of kidney function, should be interpreted with caution, given that creatinine is a breakdown product of creatine phosphate and creatine in muscle; thus the presence of high urine creatinine would be

expected during periods of high creatine consumption, and is not necessarily indicative of kidney damage (Gualano 2011).

In light of the current evidence, it is considered plausible that creatine could protect the human fetal brain against injury associated with hypoxia-ischaemia, excitotoxicity or oxidative stress, without causing harm to the fetus or the mother. It is important to assess whether maternally administered creatine (at the time of known, suspected, or potential fetal compromise) may offer fetal neuroprotection and may accordingly reduce the risk of cerebral palsy and associated impairments and disabilities arising from fetal brain injury.

Why it is important to do this review

Creatine has been shown to have neuroprotective properties (such as providing cellular energy in the absence of oxygen (Beal 2011; Wallimann 1992), demonstrating antioxidant effects (Lawler 2002; Sestili 2006), and improving cerebral blood flow following hypoxia (Prass 2006)). Animal studies have supported a fetal neuroprotective role for creatine when administered maternally (Ireland 2008; Ireland 2011). It is important to assess whether creatine, when given to pregnant women, can reduce the risk of neurological impairments and disabilities (including cerebral palsy) associated with fetal brain injury, and death, for the preterm or term fetus.

This review will complement the Cochrane review 'Melatonin for women in pregnancy for neuroprotection of the fetus' (Wilkinson 2013), which is assessing melatonin as a novel agent for preterm and/or term fetal neuroprotection, and the Cochrane reviews assessing magnesium sulphate for neuroprotection of the preterm (Doyle 2009) and term fetus (Nguyen 2013).

OBJECTIVES

To assess the effects of creatine when used for neuroprotection of the fetus.

METHODS

Criteria for considering studies for this review

Types of studies

All published, unpublished, and ongoing randomised trials and quasi-randomised trials assessing creatine for fetal neuroprotection - although none were identified. We would have included studies reported as abstracts only as well as those with full-text manuscripts. Studies using a cross-over or cluster-randomised design were not eligible for inclusion.

Types of participants

Pregnant women regardless of whether the pregnancy was single or multiple, and regardless of their gestational age. This could include, for example, trials of women with preterm or growth-restricted fetuses, with chorioamnionitis, with prelabour rupture of membranes, with pre-eclampsia, or with actual/suspected antenatal/intrapartum fetal distress.

Types of interventions

Trials where creatine was administered to pregnant women, and compared with a placebo or no treatment, or with an alternative agent aimed at providing fetal neuroprotection (e.g.

magnesium sulphate or melatonin). We also planned to include trials where creatine was administered to pregnant women where the indication for use was not fetal neuroprotection, where information had been reported on the review's pre-specified outcomes. We planned to include studies where different regimens for administration of creatine were compared. We planned to include studies regardless of the route (i.e. oral, intramuscular, or intravenous), timing, dose, and duration of creatine administration.

Types of outcome measures

Primary outcomes

We chose primary outcomes that were felt to be most representative of the clinically important measures of effectiveness and safety, including serious outcomes and adverse effects.

For the infant/child

- Death or any neurosensory disability (at latest time reported) (this combined outcome recognises the potential for competing risks of death or survival with neurological problems)
- Death (defined as all fetal, neonatal, or later death) (at latest time reported)
- Neurosensory disability (*any of cerebral palsy, blindness, deafness, developmental delay/intellectual impairment) (at latest time reported)

*Definitions

- Cerebral palsy: abnormality of tone with motor dysfunction (as diagnosed at 18 months of age or later)
- Blindness: corrected visual acuity worse than 6/60 in the better eye
- Deafness: hearing loss requiring amplification or worse
- Developmental delay/intellectual impairment: a standardised score less than minus one standard deviation (SD) below the mean (or as defined by trialists)

For the mother

- Any adverse effects severe enough to stop treatment (as defined by trialists)

Secondary outcomes

Secondary outcomes include other measures of effectiveness and safety.

For the fetus/infant

- Abnormal fetal and umbilical Doppler ultrasound study (as defined by trialists)
- Fetal death
- Neonatal death
- Gestational age at birth
- Birthweight (absolute and centile)
- Apgar score (less than seven at five minutes)
- Active resuscitation via an endotracheal tube at birth
- Use and duration of respiratory support (mechanical ventilation or continuous positive airways pressure, or both)
- Intraventricular haemorrhage (including severity - grade one to four) (as defined by trialists)
- Periventricular leukomalacia (as defined by trialists)

- Hypoxic ischaemic encephalopathy (as defined by trialists)
- Neonatal encephalopathy (as defined by trialists)
- Proven neonatal sepsis (as defined by trialists)
- Necrotising enterocolitis (as defined by trialists)
- Abnormal neurological examination (however defined by the trialists, at a point earlier than 18 months of age)

For the mother

- Side effects and serious adverse events associated with treatment (as reported by individual trialists e.g. renal dysfunction)
- Women's satisfaction with the treatment (as defined by trialists)
- Mode of birth (normal vaginal birth, operative vaginal birth, caesarean section), and indication for non-elective mode of birth

For the infant/child

- Cerebral palsy (any, and graded as severe: including children who are non-ambulant and are likely to remain so; moderate: including those children who have substantial limitation of movement; mild: including those children walking with little limitation of movement)
- Death or cerebral palsy
- Blindness
- Deafness
- Developmental delay/intellectual impairment (classified as severe: a developmental quotient or intelligence quotient less than minus three SD below the mean (or as defined by trialists); moderate: a developmental quotient or intelligence quotient from minus three SD to minus two SD below the mean (or as defined by trialists); mild: a developmental quotient or intelligence quotient from minus two SD to minus one SD below the mean (or as defined by trialists))
- Major neurosensory disability (defined as any of: moderate or severe cerebral palsy, legal blindness, neurosensory deafness requiring hearing aids, or moderate or severe developmental delay/intellectual impairment)
- Death or major neurosensory disability
- Growth assessments at childhood follow-up (weight, head circumference, length/height)

Use of health services

- Admission to intensive care unit for the mother
- Length of postnatal hospitalisation for the women
- Admission to neonatal intensive care for the infant and length of stay
- Costs of care for the mother or infant, or both

Search methods for identification of studies

Electronic searches

We contacted the Trials Search Co-ordinator to search the Cochrane Pregnancy and Childbirth Group's Trials Register (30 November 2014).

The Cochrane Pregnancy and Childbirth Group's Trials Register is maintained by the Trials Search Co-ordinator and contains trials identified from:

1. monthly searches of the Cochrane Central Register of Controlled Trials (CENTRAL);
2. weekly searches of MEDLINE (Ovid);
3. weekly searches of Embase (Ovid);
4. handsearches of 30 journals and the proceedings of major conferences;
5. weekly current awareness alerts for a further 44 journals plus monthly BioMed Central email alerts.

Details of the search strategies for CENTRAL, MEDLINE, and Embase, the list of handsearched journals and conference proceedings, and the list of journals reviewed via the current awareness service can be found in the 'Specialized Register' section within the editorial information about the [Cochrane Pregnancy and Childbirth Group](#).

Trials identified through the searching activities described above are each assigned to a review topic (or topics). The Trials Search Co-ordinator searches the register for each review using the topic list rather than keywords.

We planned not to apply any language or date restrictions.

Searching other resources

We planned to search reference lists of retrieved studies.

Data collection and analysis

See [Appendix 1](#) for methods of data collection and analysis to be used in future updates of this review.

RESULTS

Description of studies

There were no studies in the Cochrane Pregnancy and Childbirth Group's Trials Register.

Risk of bias in included studies

We found no randomised controlled trials for inclusion in the review.

Effects of interventions

We found no randomised controlled trials for inclusion in the review.

DISCUSSION

We identified no randomised controlled trials assessing the benefits and harms of creatine for women in pregnancy for neuroprotection of the fetus.

Death and neurosensory disabilities, such as cerebral palsy, are serious outcomes after a preterm or term compromised pregnancy/birth, and thus the identification of primary preventative therapies is of crucial importance. Systematic reviews show that maternal administration of corticosteroids for impending preterm birth significantly reduces the risk of neonatal death, respiratory distress, cerebroventricular haemorrhage, and necrotising enterocolitis, and clearly reduces the requirement for neonatal respiratory support and intensive care ([Roberts 2006](#)). Antenatal magnesium sulphate administration has also been shown to reduce the risk of cerebral palsy and death when administered to women

immediately prior to preterm birth (Doyle 2009). Maternal administration of the xanthine oxidase inhibitor allopurinol is under trial as a means of protecting the fetal brain from hypoxia-induced oxidative stress (Kaandorp 2010; Kaandorp 2012); and antenatal melatonin is being assessed in pilot studies, for reducing oxidative stress and brain injury in pregnancies complicated by intrauterine growth restriction (ACTRN12612000858897) and pre-eclampsia (Hobson 2013). While of potential/proven benefit, these treatments may be seen to be initiated 'late', i.e. when preterm birth is imminent or the fetus is already subjected to intrauterine hypoxia. These treatments currently require tertiary level medical care, which may restrict their use to settings with high degrees of obstetric surveillance. In the case of allopurinol, concerns have additionally been raised about its possible interference with normative and hypoxic regulation of the fetal circulation (Kane 2014). Notwithstanding the use of antenatal corticosteroids and magnesium sulphate to lower the risk of brain injury at or near birth (preterm or term), there are currently no accepted treatments that are recommended for use during the second and third trimesters of pregnancy for the purpose of preventing birth-related hypoxic-ischaemic encephalopathy.

Clinical trials have shown that long-term creatine supplementation is well tolerated, slowing the accumulation of glutamate in the brain of early-onset Huntington's Disease (Bender 2006), and without serious side effects when given over years in patients with Parkinson's Disease (Bender 2005); creatine has also been shown to improve short-term and long-term outcomes for children recovering from traumatic brain injury (Sakellaris 2006; Sakellaris 2008). Compelling evidence from recent animal studies suggests that creatine could be a simple, cheap, and effective neuroprotective strategy for the fetus when administered maternally. A recent review, summarising the experimental studies of creatine supplementation during pregnancy to date, concluded that based on current evidence, this treatment should be evaluated as a prophylactic therapy, with the potential to improve fetal and neonatal morbidity and to reduce mortality in high-risk human pregnancy (through protecting the brain, and possibly preventing damage to other organs) (Dickinson 2014). Creatine readily crosses the placenta in humans (Miller 1974) and animals (Braissant 2005; Ireland 2008), and accumulates in fetal tissues in animals (Ireland 2008). When administered maternally, creatine prevents hypoxia-induced fetal brain injury (Ireland 2011). The proposed mechanism of action is the maintenance of tissue energy levels, which prevents the activation of apoptotic and lipid peroxidation pathways (Ireland 2011). Creatine/phosphocreatine functions primarily as a spatial and temporal energy buffer, connecting sub-cellular sites of energy production with sites of energy utilisation at times of high energy demand (Wallimann 1992). In addition to yielding ATP, the dephosphorylation of creatine utilises free protons and ADP, thereby reducing the fall of intracellular pH and aiding in the stabilisation of the mitochondrial membrane potential (Wallimann 1992).

However, in the absence of randomised controlled trial data, uncertainty persists regarding the relative benefits and harms

of creatine when given to women in pregnancy for fetal neuroprotection.

AUTHORS' CONCLUSIONS

Implications for practice

As we did not identify any eligible trials for inclusion in this review, we are unable to comment on implications for practice regarding the use of creatine for women in pregnancy for neuroprotection of the fetus.

Implications for research

The available animal studies of creatine in pregnancy for fetal neuroprotection provide some insight into the potential benefits of this intervention.

Research efforts are currently being directed towards understanding creatine biology in human pregnancy, including identifying whether pregnancies in which creatine concentrations are low are associated with poorer pregnancy outcomes or vice versa. While these studies will be informative for understanding creatine biology in human pregnancy, the absence of such associations will not preclude the possibility that creatine supplementation, above concentrations normally observed toward the end of pregnancy, could provide fetal neuroprotection. Before human trials are conducted, it will be important to determine whether taking creatine during pregnancy is safe for the mother and fetus; studies in larger animals, more comparable to the human, such as primates, will be helpful in establishing this.

If the safety and efficacy of creatine treatment are established, randomised controlled trials in humans are required to provide reliable evidence about the benefits and harms of creatine for this indication. Such randomised controlled trials in human pregnancy should first compare creatine supplementation with either no intervention (ideally a placebo), or with an alternative agent aimed at fetal neuroprotection. If appropriate, these trials should then be followed by studies comparing different creatine regimens (dosage and duration of exposure). Trials must be of a high quality and adequately powered to assess the comparative effects on fetal, infant and child mortality, child morbidity including cerebral palsy and other neurosensory disabilities, maternal outcomes including adverse effects, and the use of health services.

ACKNOWLEDGEMENTS

The authors acknowledge the editorial staff and the reviewers for their helpful comments on the protocol and this review.

As part of the pre-publication editorial process, this review has been commented on by four peers (an editor and three referees who are external to the editorial team), a member of the Pregnancy and Childbirth Group's international panel of consumers and the Group's Statistical Adviser.

REFERENCES

Additional references

ACPR Group 2009

Australian Cerebral Palsy Register (ACPR) Group. Report of the Australian Cerebral Palsy Register, Birth Years 1993-2003. Cerebral Palsy Institute, 2009.

ACTRN12612000858897

ACTRN12612000858897. Melatonin to prevent brain injury in unborn growth restricted babies (MEL). <https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=362854> (accessed 10 December 2013).

Adcock 2002

Adcock KH, Nedelcu J, Loenneker T, Martin E, Wallimann T, Wagner BP. Neuroprotection of creatine supplementation in neonatal rats with transient cerebral hypoxia-ischemia. *Developmental Neuroscience* 2002;**24**(5):382-8.

Bain 2013

Bain ES, Middleton PF, Crowther CA. Maternal adverse effects of different antenatal magnesium sulphate regimens for improving maternal and infant outcomes: a systematic review. *BMC Pregnancy and Childbirth* 2013;**13**:195.

Bax 2005

Bax M, Goldstein M, Rosenbaum P, Leviton A, Paneth N, Dan B, et al. Proposed definition and classification of cerebral palsy. *Developmental Medicine and Child Neurology* 2005;**47**:571-6.

Beal 2011

Beal M. Neuroprotective effects of creatine. *Amino Acids* 2011;**40**(5):1305-13.

Bender 2005

Bender A, Auer DP, Merl T, Reilmann R, Saemann P, Yassouridis A, et al. Creatine supplementation lowers brain glutamate levels in Huntington's disease. *Journal of Neurology* 2005;**252**(1):36-41.

Bender 2006

Bender A, Koch W, Elstner M, Schombacher Y, Bender J, Moeschl M. Creatine supplementation in Parkinson disease: a placebo-controlled randomized pilot trial. *Neurology* 2006;**67**(7):1262-4.

Blair 2006

Blair E, Watson L. Epidemiology of cerebral palsy. *Seminars in Fetal and Neonatal Medicine* 2006;**11**:117-23.

Braissant 2005

Braissant O, Henry H, Villard AM, Speer O, Wallimann T, Bachmann C. Creatine synthesis and transport during rat embryogenesis: spatiotemporal expression of AGAT, GAMT and CT1. *BMC Developmental Biology* 2005;**5**:9.

Calvert 2005

Calvert, JW, Zhang JH. Pathophysiology of an hypoxic-ischemic insult during the perinatal period. *Neurological Research* 2005;**27**(3):246-60.

Cannata 2010

Cannata DJ, Ireland Z, Dickinson H, Snow RJ, Russell AP, West JM, et al. Maternal creatine supplementation from mid-pregnancy protects the diaphragm of the newborn spiny mouse from intrapartum hypoxia-induced damage. *Pediatric Research* 2010;**68**(5):393-8.

Dickinson 2013

Dickinson H, Ireland ZJ, Larosa DA, O'Connell BA, Ellery S, Snow R, et al. Maternal dietary creatine supplementation does not alter the capacity for creatine synthesis in the newborn spiny mouse. *Reproductive Sciences* 2013;**20**(9):1096-102.

Dickinson 2014

Dickinson H, Ellery S, Ireland Z, Larosa D, Snow R, Walker DW. Creatine supplementation during pregnancy: summary of experimental studies suggesting a treatment to improve fetal and neonatal morbidity and reduce mortality in high-risk human pregnancy. *BMC Pregnancy and Childbirth* 2014;**14**:150.

Dilenge 2001

Dilenge M, Majnemer A, Shevell M. Long-term developmental outcome of asphyxiated term neonates. *Journal of Child Neurology* 2001;**16**:781-92.

Doyle 2009

Doyle LW, Crowther CA, Middleton P, Marret S, Rouse D. Magnesium sulphate for women at risk of preterm birth for neuroprotection of the fetus. *Cochrane Database of Systematic Reviews* 2009, Issue 1. [DOI: [10.1002/14651858.CD004661.pub3](https://doi.org/10.1002/14651858.CD004661.pub3)]

du Plessis 2002

du Plessis AJ, Volpe JJ. Perinatal brain injury in the preterm and term newborn. *Current Opinion in Neurology* 2002;**15**:151-7.

Ellery 2013

Ellery SJ, Ireland Z, Kett MM, Snow R, Walker DW, Dickinson H. Creatine pretreatment prevents birth asphyxia-induced injury of the newborn spiny mouse kidney. *Pediatric Research* 2013;**73**(2):201-8.

Gates 2004

Gates S, Brocklehurst P. How should trials recruiting women with multiple pregnancies be analysed?. *British Journal of Obstetrics and Gynaecology* 2004;**111**:213-9.

Gualano 2011

Gualano B, de Salles Painelli V, Roschel H, Lugaresi R, Dorea E, Artioli GG. Creatine supplementation does not impair kidney function in type 2 diabetic patients: a randomized, double-blind, placebo-controlled, clinical trial. *European Journal of Applied Physiology* 2011;**111**(5):749-56.

Higgins 2011

Higgins JPT, Green S (editors). Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0 [updated March 2011]. The Cochrane Collaboration, 2011. Available from www.cochrane-handbook.org.

Hobson 2013

Hobson SR, Lim R, Gardiner EE, Alers NO, Wallace EM. Phase I pilot clinical trial of antenatal maternally administered melatonin to decrease the level of oxidative stress in human pregnancies affected by pre-eclampsia (PAMPR): study protocol. *BMJ Open* 2013;**3**:e003788.

Ireland 2008

Ireland Z, Dickinson H, Snow R, Walker DW. Maternal creatine: does it reach the fetus and improve survival after an acute hypoxic episode in the spiny mouse (*Acomys cahirinus*)?. *American Journal of Obstetrics and Gynecology* 2008;**198**(4):431e1-6.

Ireland 2011

Ireland Z, Castillo-Melendez M, Dickinson H, Snow R, Walker DW. A maternal diet supplemented with creatine from mid-pregnancy protects the newborn spiny mouse brain from birth hypoxia. *Neuroscience* 2011;**194**:372-9.

Jacobsson 2002

Jacobsson B, Hagberg G, Hagberg B, Ladfors L, Niklasson A, Hagberg H. Cerebral palsy in preterm infants: a population-based case-control study of antenatal and intrapartum risk factors. *Acta Paediatrica* 2002;**91**:946-51.

Jensen 2003

Jensen A, Garnier Y, Middelani J, Berger R. Perinatal brain damage—from pathophysiology to prevention. *European Journal of Obstetrics, Gynecology, and Reproductive Biology* 2003;**110**:S70-9.

Johnston 2000

Johnston MV, Trescher WH, Ishida A, Nakajima W. Novel treatments after experimental brain injury. *Seminars in Neonatology* 2000;**5**:75-86.

Kaandorp 2010

Kaandorp JJ, Benders MJ, Rademaker CM, Torrance HL, Oudijk MA, de Haan TR, et al. Antenatal allopurinol for reduction of birth asphyxia induced brain damage (ALLO-Trial); a randomized double blind placebo controlled multicenter study. *BMC Pregnancy and Childbirth* 2010;**10**:8.

Kaandorp 2012

Kaandorp JJ, van Bel F, Veen S, Derks JB, Groenendaal F, Rijken M, et al. Long-term neuroprotective effects of allopurinol after moderate perinatal asphyxia: follow-up of two randomised controlled trials. *Archives of Disease in Childhood. Fetal and Neonatal Edition* 2012;**97**(3):F162-166.

Kane 2014

Kane AD, Hansell JA, Herrera EA, Allison BJ, Niu Y, Brain KL, et al. Xanthine oxidase and the fetal cardiovascular defence to

hypoxia in late gestation ovine pregnancy. *Journal of Physiology* 2014;**592**(Pt 3):4765-89.

Larroque 2003

Larroque B, Marret S, Ancel PY, Arnaud C, Marpeau L, Supernant K, et al. White matter damage and intraventricular hemorrhage in very preterm infants: the EPIPAGE study. *Journal of Pediatrics* 2003;**143**(4):477-83.

Lawler 2002

Lawler JM, Barnes WS, Wu G, Song W, Demaree S. Direct antioxidant properties of creatine. *Biochemical and Biophysical Research Communications* 2002;**290**(1):47-52.

Marret 2007

Marret S, Doyle LW, Crowther CA, Middleton P. Antenatal magnesium sulphate neuroprotection in the preterm infant. *Seminars in Fetal & Neonatal Medicine* 2007;**12**(4):311-7.

McIntyre 2013

McIntyre S, Taitz D, Keogh J, Goldsmith S, Badawi N, Blair E. A systematic review of risk factors for cerebral palsy in children born at term in developed countries. *Developmental Medicine and Child Neurology* 2013;**55**:499-508.

Miller 1974

Miller R, Davis BM, Brent RL, Koszalka TR. Transport of creatine in the human placenta. *Pharmacologist* 1974;**16**(2):305.

Nguyen 2013

Nguyen TMN, Crowther CA, Wilkinson D, Bain E. Magnesium sulphate for women at term for neuroprotection of the fetus. *Cochrane Database of Systematic Reviews* 2013, Issue 2. [DOI: [10.1002/14651858.CD009395.pub2](https://doi.org/10.1002/14651858.CD009395.pub2)]

Novak 2012

Novak I, Hines M, Goldsmith S, Barclay R. Clinical prognostic messages from a systematic review of cerebral palsy. *Paediatrics* 2012;**130**(5):e1285-312.

Pastula 2012

Pastula DM, Moore DH, Bedlack RS. Creatine for amyotrophic lateral sclerosis/motor neuron disease. *Cochrane Database of Systematic Reviews* 2012, Issue 12. [DOI: [10.1002/14651858.CD005225.pub3](https://doi.org/10.1002/14651858.CD005225.pub3)]

Prass 2006

Prass K, Royl G, Lindauer U, Freyer D, Megow D, Dirnagl U, et al. Improved reperfusion and neuroprotection by creatine in a mouse model of stroke. *Journal of Cerebral Blood Flow and Metabolism* 2006;**27**(3):452-9.

Rees 2011

Rees S, Harding R, Walker D. The biological basis of injury and neuroprotection in the fetal and neonatal brain. *International Journal of Developmental Neuroscience* 2011;**29**(6):551-63.

RevMan 2014 [Computer program]

The Nordic Cochrane Centre, The Cochrane Collaboration. Review Manager (RevMan). Version 5.3. Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration, 2014.

Roberts 2006

Roberts D, Dalziel S. Antenatal corticosteroids for accelerating fetal lung maturation for women at risk of preterm birth. *Cochrane Database of Systematic Reviews* 2006, Issue 3. [DOI: [10.1002/14651858.CD004454.pub2](https://doi.org/10.1002/14651858.CD004454.pub2)]

Sakellaris 2006

Sakellaris G, Kotsiou M, Tamiolaki M, Kalostos G, Tsapaki E, Spanaki M, et al. Prevention of complications related to traumatic brain injury in children and adolescents with creatine administration: an open label randomized pilot study. *Journal of Trauma* 2006;**61**(2):322-9.

Sakellaris 2008

Sakellaris G, Nasis G, Kotsiou M, Tamiolaki M, Charissis G, Evangelou A. Prevention of traumatic headache, dizziness and fatigue with creatine administration. A pilot study. *Acta Paediatrica* 2008;**97**(1):31-4.

Sestili 2006

Sestili P, Martinelli C, Bravi G, Piccoli G, Curci R, Battistelli M, et al. Creatine supplementation affords cytoprotection in oxidatively injured cultured mammalian cells via direct antioxidant activity. *Free Radical Biology and Medicine* 2006;**40**(5):837-49.

Sullivan 2000

Sullivan PG, Geiger JD, Mattson MP, Scheff SW. Dietary supplement creatine protects against traumatic brain injury. *Annals of Neurology* 2000;**48**(5):723-9.

Volpe 2001

Volpe JJ. Perinatal brain injury: from pathogenesis to neuroprotection. *Mental Retardation and Developmental Disabilities* 2001;**7**:56-64.

Wallimann 1992

Wallimann TWM, Brdiczka D, Nicolay K, Eppenberger H. Intracellular compartmentation, structure and function of creatine kinase isoenzymes in tissues with high and fluctuating energy demands: the "phosphocreatine circuit" for cellular energy homeostasis. *Biochemical Journal* 1992;**281**(Pt 1):21-40.

Wallimann 2011

Wallimann T, Tokarska-Schlattner M, Schlattner U. The creatine kinase system and pleiotropic effects of creatine. *Amino Acids* 2011;**40**(5):1271-96.

Wang 2012

Wang J, Xiao Y, Chen S, Zhong C, Luo M, Luo H. Creatine for Parkinson's disease. *Cochrane Database of Systematic Reviews* 2012, Issue 2. [DOI: [10.1002/14651858.CD009646](https://doi.org/10.1002/14651858.CD009646)]

Wilkinson 2013

Wilkinson D, Bain E, Wallace E. Melatonin for women in pregnancy for neuroprotection of the fetus. *Cochrane Database of Systematic Reviews* 2013, Issue 5. [DOI: [10.1002/14651858.CD010527](https://doi.org/10.1002/14651858.CD010527)]

Wu 2003

Wu Y, Escobar G, Grether J, Croen L, Greene J, Newman T. Chorioamnionitis and cerebral palsy in term and near-term infants. *JAMA* 2003;**290**(20):2677-84.

Wyss 2002

Wyss M, Schulze A. Health implications of creatine: can oral creatine supplementation protect against neurological and atherosclerotic disease?. *Neuroscience* 2002;**112**(2):243-60.

Zhu 2004

Zhu S, Li M, Figueroa BE, Liu A, Stavrovskaya IG, Pasinelli P, et al. Prophylactic creatine administration mediates neuroprotection in cerebral ischemia in mice. *Journal of Neuroscience* 2004;**24**(26):5909-12.

APPENDICES

Appendix 1. Methods of data collection and analysis to be used in future updates of this review

Selection of studies

At least two review authors will independently assess for inclusion all the potential studies we identify as a result of the search strategy. We will resolve any disagreement through discussion or, if required, we will consult a third review author.

Data extraction and management

We will design a form to extract data. For eligible studies, at least two review authors will extract the data using the agreed form. We will resolve discrepancies through discussion or, if required, we will consult a third review author. We will enter data into the Review Manager software ([RevMan 2014](#)) and check for accuracy.

When information regarding any of the above is unclear, we will attempt to contact authors of the original reports to provide further details.

Assessment of risk of bias in included studies

At least two review authors will independently assess risk of bias for each study using the criteria outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* ([Higgins 2011](#)). We will resolve any disagreement by discussion or by involving a third assessor.

(1) Random sequence generation (checking for possible selection bias)

We will describe for each included study the method used to generate the allocation sequence in sufficient detail to allow an assessment of whether it should produce comparable groups.

We will assess the method as:

- low risk of bias (any truly random process, e.g. random number table; computer random number generator);
- high risk of bias (any non-random process, e.g. odd or even date of birth; hospital or clinic record number);
- unclear risk of bias.

(2) Allocation concealment (checking for possible selection bias)

We will describe for each included study the method used to conceal allocation to interventions prior to assignment and will assess whether intervention allocation could have been foreseen in advance of, or during recruitment, or changed after assignment.

We will assess the methods as:

- low risk of bias (e.g. telephone or central randomisation; consecutively numbered, sealed, opaque envelopes);
- high risk of bias (open random allocation; unsealed or non-opaque envelopes; alternation; date of birth);
- unclear risk of bias.

(3.1) Blinding of participants and personnel (checking for possible performance bias)

We will describe for each included study the methods used, if any, to blind study participants and personnel from knowledge of which intervention a participant received. We will consider that studies are at low risk of bias if they were blinded, or if we judge that the lack of blinding would be unlikely to affect results. We will assess blinding separately for different outcomes or classes of outcomes.

We will assess the methods as:

- low, high, or unclear risk of bias for participants;
- low, high, or unclear risk of bias for personnel.

(3.2) Blinding of outcome assessment (checking for possible detection bias)

We will describe for each included study the methods used, if any, to blind outcome assessors from knowledge of which intervention a participant received. We will assess blinding separately for different outcomes or classes of outcomes.

We will assess methods used to blind outcome assessment as:

- low, high, or unclear risk of bias.

(4) Incomplete outcome data (checking for possible attrition bias due to the amount, nature, and handling of incomplete outcome data)

We will describe for each included study, and for each outcome or class of outcomes, the completeness of data including attrition and exclusions from the analysis. We will state whether attrition and exclusions were reported and the numbers included in the analysis at each stage (compared with the total randomised participants), reasons for attrition or exclusion where reported, and whether missing data were balanced across groups or were related to outcomes. Where sufficient information is reported, or can be supplied by the trial authors, we will re-include missing data in the analyses which we undertake.

We will assess methods as:

- low risk of bias (e.g. no missing outcome data; missing outcome data balanced across groups);
- high risk of bias (e.g. numbers or reasons for missing data imbalanced across groups; 'as treated' analysis done with substantial departure of intervention received from that assigned at randomisation);
- unclear risk of bias.

(5) Selective reporting (checking for reporting bias)

We will describe for each included study how we investigated the possibility of selective outcome reporting bias and what we found.

We will assess the methods as:

- low risk of bias (where it is clear that all of the study's pre-specified outcomes and all expected outcomes of interest to the review have been reported);

- high risk of bias (where not all the study's pre-specified outcomes have been reported; one or more reported primary outcomes were not pre-specified; outcomes of interest are reported incompletely and so cannot be used; study fails to include results of a key outcome that would have been expected to have been reported);
- unclear risk of bias.

(6) Other bias (checking for bias due to problems not covered by (1) to (5) above)

We will describe for each included study any important concerns we have about other possible sources of bias.

We will assess whether each study was free of other problems that could put it at risk of bias:

- low risk of other bias;
- high risk of other bias;
- unclear whether there is risk of other bias.

(7) Overall risk of bias

We will make explicit judgements about whether studies are at high risk of bias, according to the criteria given in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011). With reference to (1) to (6) above, we will assess the likely magnitude and direction of the bias and whether we consider it is likely to impact on the findings. We will explore the impact of the level of bias through undertaking sensitivity analyses - see Sensitivity analysis.

Measures of treatment effect

Dichotomous data

For dichotomous data, we will present results as summary risk ratio with 95% confidence intervals.

Continuous data

For continuous data, we will use the mean difference if outcomes are measured in the same way between trials. We will use the standardised mean difference to combine trials that measure the same outcome, but use different methods.

Unit of analysis issues

Cluster-randomised trials

We consider cluster-randomised trials as inappropriate for this research question.

Cross-over trials

We consider cross-over trials as inappropriate for this research question.

Multiple pregnancies

As infants from multiple pregnancies are not independent, we plan to use cluster trial methods in the analyses, where the data allow, and where multiples make up a substantial proportion of the trial population, to account for non-independence of variables (Gates 2004).

Multi-armed studies

If multi-armed studies are included in the review, we plan to combine groups where appropriate in order to create a single pair-wise comparison (e.g. creatine versus alternative neuroprotective treatments).

If an included trial has an intervention arm that is not relevant to the review question, we will comment on this in the table of 'Characteristics of included studies', and include in the review only the intervention and control groups that meet the eligibility criteria.

Dealing with missing data

For included studies, we will note levels of attrition. We will explore the impact of including studies with high levels of missing data in the overall assessment of treatment effect by using sensitivity analysis.

For all outcomes, we will carry out analyses, as far as possible, on an intention-to-treat basis, i.e. we will attempt to include all participants randomised to each group in the analyses, and we will analyse all participants in the group to which they were allocated, regardless of whether or not they received the allocated intervention. The denominator for each outcome in each trial will be the number randomised minus any participants whose outcomes are known to be missing.

Assessment of heterogeneity

We will assess statistical heterogeneity in each meta-analysis using the T^2 , I^2 , and Chi^2 statistics. We will regard heterogeneity as substantial if an I^2 is greater than 30% and either a T^2 is greater than zero, or there is a low P value (less than 0.10) in the Chi^2 test for heterogeneity.

Assessment of reporting biases

If there are 10 or more studies in the meta-analysis, we will investigate reporting biases (such as publication bias) using funnel plots. We will assess funnel plot asymmetry visually. If asymmetry is suggested by a visual assessment, we will perform exploratory analyses to investigate it.

Data synthesis

We will carry out statistical analysis using the Review Manager software ([RevMan 2014](#)). We will use fixed-effect meta-analysis for combining data where it is reasonable to assume that studies are estimating the same underlying treatment effect: i.e. where trials are examining the same intervention, and the trials' populations and methods are judged sufficiently similar. If there is clinical heterogeneity sufficient to expect that the underlying treatment effects differ between trials, or if substantial statistical heterogeneity is detected, we will use random-effects meta-analysis to produce an overall summary, if an average treatment effect across trials is considered clinically meaningful. We will treat the random-effects summary as the average range of possible treatment effects and we will discuss the clinical implications of treatment effects differing between trials. If the average treatment effect is not clinically meaningful, we will not combine trials.

If we use random-effects analyses, we will present the results as the average treatment effect with 95% confidence intervals, and the estimates of T^2 and I^2 .

Subgroup analysis and investigation of heterogeneity

We will perform separate comparisons for those trials comparing creatine with no treatment or a placebo, and those comparing creatine with an alternative neuroprotective agent.

If we identify substantial heterogeneity, we will investigate it using subgroup analyses and sensitivity analyses. We will consider whether an overall summary is meaningful, and, if it is, use random-effects analysis to produce it.

Maternal characteristics, and characteristics of the intervention, are likely to affect health outcomes. We will carry out subgroup analyses, if sufficient data are available, based on:

- gestational age at which the woman commenced creatine treatment (e.g. < 26 weeks versus 26 to < 28 weeks versus 28 to < 30 weeks versus 30 to < 32 weeks versus 32 to < 34 weeks versus 34 to < 37 weeks versus 37 weeks and over);
- reasons the mother was considered for creatine treatment (e.g. preterm versus growth-restricted fetus versus prolonged prelabour rupture of membrane versus increased risk of chorioamnionitis versus pre-eclampsia/eclampsia versus increased risk of perinatal asphyxia versus other);
- total daily dose of creatine administered (e.g. low (≤ 1 g per day) versus moderate (> 1 g to ≤ 5 g per day) versus high (> 5 g per day));
- mode of administration (e.g. intramuscular versus intravenous versus oral);
- number of babies *in utero* (e.g. singleton versus multiple).

We will use primary outcomes in subgroup analyses.

We will assess subgroup differences by interaction tests available within [RevMan 2014](#). We will report the results of subgroup analyses quoting the χ^2 statistic and P value, and the interaction test I^2 value.

Sensitivity analysis

We will carry out sensitivity analysis to explore the effects of trial quality assessed by allocation concealment and random sequence generation (considering selection bias), by omitting studies rated as 'high risk of bias' or 'unclear risk of bias' for these components. We will restrict this to the primary outcomes.

CONTRIBUTIONS OF AUTHORS

Emily Bain drafted the first version of the protocol. Hayley Dickinson, David Walker, Dominic Wilkinson, Philippa Middleton, and Caroline Crowther made comments and contributed to the subsequent drafts of the protocol.

Hayley Dickinson drafted the first version of the review, with David Walker, Emily Bain, Dominic Wilkinson, Philippa Middleton, and Caroline Crowther commenting on the draft, and contributing to the final version.

DECLARATIONS OF INTEREST

Hayley Dickinson: None known.

Emily Bain: None known.

Dominic Wilkinson: None known.

Philippa Middleton: None known.

Caroline A Crowther: None known.

David W Walker: None known.

SOURCES OF SUPPORT

Internal sources

- ARCH: Australian Research Centre for Health of Women and Babies, Robinson Research Institute, The University of Adelaide, Australia.

External sources

- National Health and Medical Research Council, Australia.

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

None.

INDEX TERMS

Medical Subject Headings (MeSH)

Brain Diseases [*prevention & control]; Creatine [*therapeutic use]; Fetal Diseases [*prevention & control]; Neuroprotective Agents [*therapeutic use]

MeSH check words

Female; Humans; Pregnancy